

Planning and Reinforcement Learning in AI Systems

(PARLAI)

Multi-agent AI (part 2)

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Multi-agent RL

From Multi-Agent Planning to Multi-Agent RL

Multi-agent RL

Self-Interest and
Market Based
Approaches

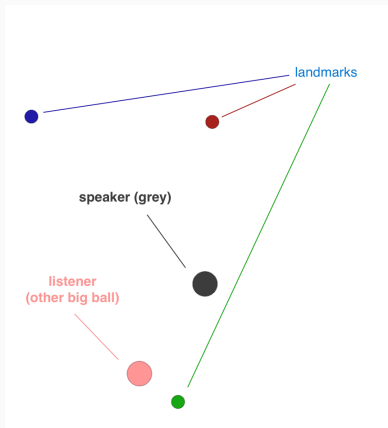
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Conclusions

Multi-agent planning usually assumes that a multi-agent MDP model is available:

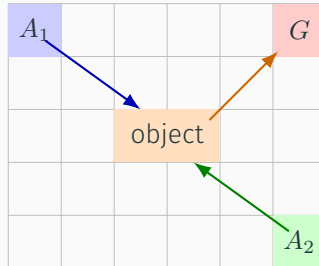
Multi-agent RL asks what happens when agents must **learn from interaction**: unknown dynamics, unknown rewards, unknown teammates, changing behavior.

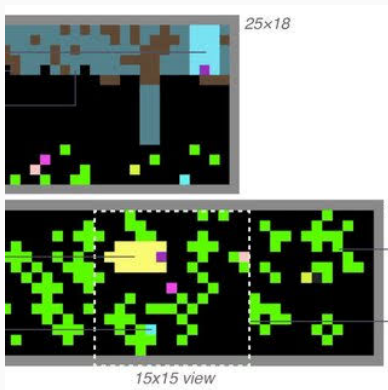


Two agents must coordinate to reach goals or move an object.

Two agents must coordinate to reach goals or move an object. A good joint policy may require:

- role assignment,
- avoiding collisions,
- waiting for another agent,
- sacrificing local reward for team reward.





Multi-agent Reinforcement Learning in Sequential Social Dilemmas

Joel Z. Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, Thore Graepel

Why Multi-Agent RL Is Harder than Single-Agent RL

In single-agent RL, the environment is usually treated as stationary.

In MARL, from the perspective of one agent:

$$P(s' \mid s, a_i) = \sum_{a_{-i}} P(s' \mid s, a_i, a_{-i}) \pi_{-i}(a_{-i} \mid s).$$

Main difficulty

Each agent is learning in an environment whose behavior depends on other learners.

If the other agents change their policies during learning, then this effective transition model changes too.

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Joint Policies and Objectives

Each agent has a policy:

$$\pi_i(a_i \mid o_i)$$

where o_i may be the full state, a local observation, or a history.

The joint policy is:

$$\Pi(\mathbf{a} \mid \mathbf{o}) = \prod_{i=1}^n \pi_i(a_i \mid o_i).$$

Agent i maximizes:

$$V_i(\Pi) = \mathbb{E}_{\Pi} \left[\sum_{t=0}^{\infty} \gamma^t r_i(s_t, \mathbf{a}_t) \right].$$

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Reward Structures

The reward structure determines the type of multi-agent problem.

Setting	Reward relation	Example
Fully cooperative	$r_1 = \dots = r_n$	robot team, traffic control
Competitive	$r_1 = -r_2$	games, pursuit-evasion
Mixed-motive	rewards partly aligned	markets, negotiation, routing

The same interaction types appear in multi-agent planning, but MARL must also learn the behavior model and the value of actions from data.

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The simplest approach is to let each agent run a single-agent RL algorithm.

$$Q_i(s, a_i) \leftarrow Q_i(s, a_i) + \alpha \left[r_i + \gamma \max_{a'_i} Q_i(s', a'_i) - Q_i(s, a_i) \right].$$

Why it is attractive

Simple, decentralized, and scalable.

Why it can fail

The update ignores the actions and learning dynamics of other agents, so the apparent environment is non-stationary.

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Joint-Action Learners

An alternative is to learn values over joint actions:

$$Q_i(s, a_1, \dots, a_n).$$

This makes the dependence on other agents explicit.

Benefit

The value function can represent coordination patterns such as:

$$Q(s, \text{lift}, \text{lift}) \gg Q(s, \text{lift}, \text{wait}).$$

Cost

The joint action space grows exponentially:

$$|\mathcal{A}_1 \times \dots \times \mathcal{A}_n| = \prod_i |\mathcal{A}_i|.$$

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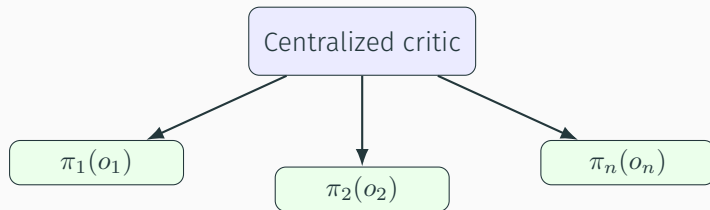
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Centralized Training, Decentralized Execution

A common compromise is **centralized training, decentralized execution** (CTDE).



Idea

During training, use global information such as the full state and joint actions. During execution, each agent acts only from its local observation.

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Multi-Agent Actor–Critic

In centralized actor–critic methods, the critic may observe the joint information:

$$Q_i^\pi(s, a_1, \dots, a_n).$$

Each actor remains local:

$$\pi_i(a_i \mid o_i).$$

A typical policy-gradient update has the form:

$$\nabla_{\theta_i} J_i = \mathbb{E} [\nabla_{\theta_i} \log \pi_i(a_i \mid o_i) A_i(s, \mathbf{a})].$$

Intuition

The critic helps agent i evaluate whether its action was useful in the context of what the other agents did.

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Value Decomposition for Cooperative MARL

For fully cooperative tasks, all agents share a team value:

$$Q_{\text{tot}}(s, \mathbf{a}).$$

Value-decomposition methods approximate this joint value using local utilities.

$$Q_{\text{tot}}(s, \mathbf{a}) \approx \sum_{i=1}^n Q_i(o_i, a_i).$$

More expressive methods learn a mixing function:

$$Q_{\text{tot}} = f_{\text{mix}}(Q_1, \dots, Q_n, s).$$

The team can train with a global objective while each agent still chooses actions using local information.

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A central question in cooperative MARL is:

Which agent caused the team success or failure?

With a shared reward, every agent receives the same signal:

$$r_1 = \dots = r_n = r_{\text{team}}.$$

This can make learning slow because one agent cannot easily tell whether its own action helped.

Common approaches

Difference rewards, counterfactual baselines, centralized critics, and value decomposition.

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Communication and Coordination

Agents may coordinate through explicit communication:

$$\pi_i(a_i, m_i \mid o_i),$$

where m_i is a message sent to other agents.

What communication can provide

- share local observations,
- announce intentions,
- request help,
- form roles or plans online.

Main challenge

Communication must be useful, cheap, robust, and interpretable enough for deployment.

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In many MARL domains, no agent sees the full state.

$$o_i \sim O_i(o_i \mid s).$$

Policies may depend on observation histories:

$$\pi_i(a_i \mid h_i^t), \quad h_i^t = (o_i^0, a_i^0, \dots, o_i^t).$$

This is the learning analogue of decentralized partially observable planning. The same information-structure problem appears, but the model and policies are learned from data.

In zero-sum games, one agent's gain is another agent's loss:

$$r_1(s, \mathbf{a}) = -r_2(s, \mathbf{a}).$$

The objective is often written as a minimax problem:

$$\max_{\pi_1} \min_{\pi_2} \mathbb{E}_{\pi_1, \pi_2} \left[\sum_t \gamma^t r_1(s_t, a_{1,t}, a_{2,t}) \right].$$

Important concept

The goal is not just high return against one opponent, but robust performance against strategic adaptation.

Many real systems are neither purely cooperative nor purely competitive.

Examples:

- autonomous vehicles sharing roads,
- energy consumers and aggregators,
- market participants,
- human-AI teams,
- robots sharing workspace and resources.

Key issue

The learned policy must reason about incentives, coordination, competition, and possibly norms or constraints.

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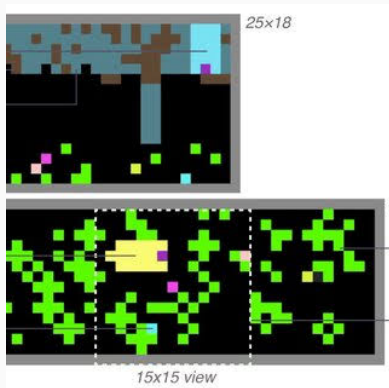
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Mixed-Motive MARL



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Common MARL Algorithm Families

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Family	Typical idea	Best fit
Independent RL	one learner per agent	simple decentralized settings
Joint-action RL	learn over joint actions	small teams
CTDE actor-critic	centralized critic, local actors	partial observability
Value decomposition	factor team value	cooperative tasks
Self-play	train against evolving opponents	competitive games
Population-based training	train with diverse partners	generalization

Conclusion

Multi-agent planning gives us formal tools for reasoning about joint action, coordination, and strategic interaction.

Multi-agent RL adds the problem of learning these behaviors from experience.

Three recurring themes

- **Non-stationarity:** other agents are part of the environment and may learn.
- **Credit assignment:** team reward does not say who caused success.
- **Scalability:** joint states and joint actions grow quickly.

MARL is planning under interaction, uncertainty, and learning.

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Self-Interest and Market Based Approaches

Accounting for Self Interested Agents

- In **cooperative** multiagent systems we can treat the collection of agents as a single agent with joint actions - “one mind.”
- When agents are self-interested or have limited communication choices made separately may be jointly suboptimal.
- **Coordination** can be based on
 - **Communication**
 - e.g., agents might communicate in order to determine task allocation.
 - **Convention (social laws)**
 - e.g., rules might be imposed by the system designed so that optimal joint action is assured.
 - **Learning**
 - e.g., learning by repeated interaction.



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Accounting for Self Interested Agents

When agents are self interested, we can allow agent to negotiate by establishing some negotiation protocol (rules of the game).

- What is a deal?
- What is the utility of each agent?
- What is the **conflict** (default) deal ?
- Negotiation strategy - how an agent should act given a set of rules ?



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The ALLIANCE Architecture - Parker 1998

A framework for multi-robot cooperation for small- to medium-sized teams of heterogeneous robots performing missions composed of independent subtasks that may have ordering dependencies.

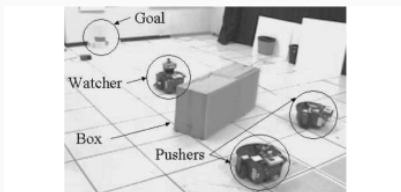


Fig. 3. Our experimental box-pushing setup. The task is for the pushers to move the box to the goal with the help of the watcher. (Image 1 of 3 taken from an experimental trial; see Figs. 4 and 5).

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The ALLIANCE Architecture - Parker 1998

Assumptions:

- 1 Robots detect the effect of their own actions, with some probability greater than 0.
- 2 Robots can detect the actions of other team members with some probability greater than 0; these actions may be detected through any available means, including explicit broadcast communication.
- 3 Robots on the team **do not lie** and are not intentionally adversarial.
- 4 The communications medium is not guaranteed to be available.
- 5 The robots do not possess perfect sensors and effectors.
- 6 Any of the robot subsystems can fail, with some probability greater than 0.
- 7 If a robot fails, it cannot necessarily communicate its failure to its teammates.
- 8 A centralized store of complete world knowledge is not available

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The ALLIANCE Architecture - Parker 1998



An architecture for task allocation in heterogeneous robot teams
- introduced behavior sets for action selection without explicit negotiations between robots.

The level of motivation is calculated based on a few factors:

- The previous motivation level
- The rate of impatience
- Is the behavior set needed, i.e. is the task required?
- Is the robot already working on a different task?
- Is a different robot already working on this task?

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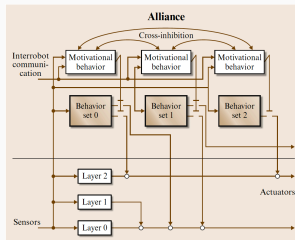
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The ALLIANCE Architecture

The motivation level keeps growing, until a certain threshold is crossed and then it can be said that a robot has selected an action.

The motivation level returns to zero in one of three cases:

- 1 The sensory feedback indicated that the behavior set is no longer needed.
- 2 Another behavior set is already activated.
- 3 Another robot has taken the task for the first time.



L-ALLIANCE: Robots can adopt the rate of impatience and acquiescence depending on the quality with which the robot can accomplish a given task. As a result, robots that were better in certain tasks are more likely to choose those tasks in the future.

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Market-Based Approaches for Multi-Agent Task Allocation

- Self-interested agents have the primary goal of maximizing individual profits.
- Relevant also as a modeling tool: revenue/cost models and rules of engagement will be designed so that maximizing individual profit has the benevolent effect of moving the team toward the globally optimal solution.
- This architecture will inherit the flexibility of market-based approaches in allowing cooperation and competition to emerge opportunistically.

A Market Approach to Multirobot Coordination. Dias et al. 2001

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Auctions for Multi-Agent Task Allocation

- Auctions are most common mechanisms used in market-based approaches.
- In an auction, a set of items is offered by an auctioneer in an **announcement phase**, and the participants can make an offer for these items by submitting bids to the auctioneer.
- Once all bids are received or a prespecified deadline has passed, the auction is then cleared in the **winner determination phase** by the auctioneer who decides which items to award and to whom.



A Market Approach to Multirobot Coordination. Dias et al. 2001

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Auctions for Multi-Agent Task Allocation

- Market-based approaches, and auctions in particular, distribute planning required for task allocation through the auction process: each agent or group of agents locally plans the achievement of the offered tasks, computes its costs, and encapsulates the costs in its bids.
- Although many approaches to task allocation assume that a list of primitive or simple tasks is input to the system, a complex mission is often more naturally described at a higher level of abstraction.
- By incorporating bids at multiple levels of abstraction, both the costs of allocations and plans can be compared in a single auction mechanism.



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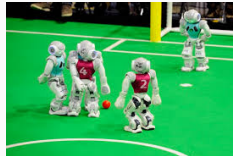
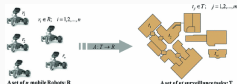
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Conclusions

Auctions for Multi-Agent Task Allocation

- Global (near) optimality via local computations
- Task reallocation is possible by introducing peer-to-peer auction
 - In this case, there is some initial allocation, and any agent on the team is capable of holding auctions in order to reallocate tasks to agents that are better suited to perform them.
- Accounting for heterogeneous teams of agents with changing needs.



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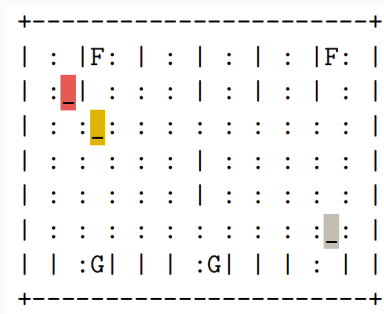
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Auctions for Solving The Multi-Taxi Problem



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A Market Economy Perspective on Reinforcement Learning- Chang et al. 2022

- A framework for societal decision-making, a perspective on reinforcement learning through the lens of a self-organizing society of primitive agents.
- From corporations to organisms, many large-scale systems in our world are composed of smaller individual autonomous components, whose collective function serve a larger objective than the objective of any individual component alone.



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Conclusions

A Market Economy Perspective on Reinforcement Learning- Chang et al. 2022

- At the core of building complex learning systems at multiple levels of abstraction is to understand the mechanisms that bind consecutive levels together.
- In the context of learning for decision-making, this means to define three ingredients:
 - A framework for expressing the encapsulation of a society of primitive agents as a super-agent
 - An incentive mechanism that guarantees the optimal solution for the super-agent's decision problem emerges as a consequence of the primitives optimizing their individual decision problems
 - A learning algorithm for implicitly training the super-agent by directly training the primitives

The incentive mechanism is the abstraction barrier that connects the optimization problems of the primitive agents from the optimization problem of the society as a super-agent.

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A Market Economy Perspective on Reinforcement Learning- Chang et al. 2022

- We consider a restricted scenario that builds upon existing frameworks familiar to us, Markov decision processes (MDP).
- Normally, the objective of the learner is to maximize the expected return of the MDP.
- With the standard monolithic decision-making framework all the learnable parameters are globally coupled together under a single objective.
- The monolithic decision-making framework views reinforcement learning from the perspective of a command economy, in which all production — the transformation of past states s_t into future states s_{t+1} — and wealth distribution — the credit assignment of reward signals to parameters — derive directly from single central authority — the MDP objective.

<https://arxiv.org/abs/2007.02382> <https://arxiv.org/abs/2007.02382>

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A Market Economy Perspective on Reinforcement Learning- Chang et al. 2022

- We can also view reinforcement learning from the perspective of a **market economy**, in which production and wealth distribution are governed by the economic transactions between actions that buy and sell states to each other.
- Rather than being passively chosen by a global policy as in the monolithic framework, actions are primitive agents that actively choose themselves when to activate in the environment by bidding in an auction to transform the state s_t to the next state s_{t+1} .
- This is a societal decision-making framework because these actions form a society of primitive agents that themselves seek to maximize their auction utility at each state: In other words, the society of primitive agents form a super-agent that solves the MDP as a consequence of the primitive agents' optimal auction strategies.

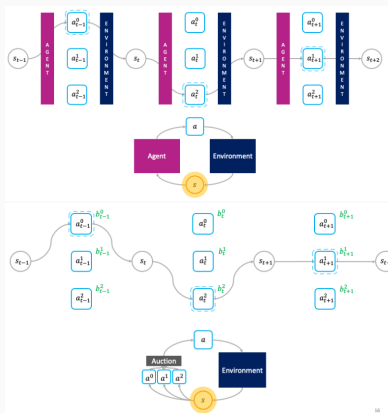
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Why Recognize Intent?

Autonomous agents rarely act alone

To collaborate effectively, an intelligent system must anticipate what others are trying to achieve before they explicitly communicate it.

- Assist humans before they ask for help.
- Coordinate with teammates.
- Detect unexpected or unsafe behavior.
- Adapt plans as others change their intentions.

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Why Recognize Intent?

Examples

- A household robot brings the correct tool after observing the user's actions.
- An autonomous vehicle predicts that a pedestrian intends to cross.
- A warehouse robot avoids blocking a teammate's planned path.
- An energy management system anticipates future electricity demand.



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Conclusions

Goal and Plan Recognition

Problem Definition

Given a sequence of observations of an agent's behavior, infer:

- the agent's underlying **goal**,
- the **plan** it is executing, or
- its future actions.

Inputs:

- Partial observations of actions or states
- Domain model (actions, dynamics, costs)
- Candidate goals (optional)

Outputs:

- Most likely goal
- Current plan or policy
- Predicted future behavior

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Goal Recognition

A goal recognition setting



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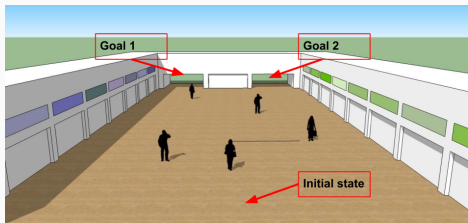
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Goal Recognition

A goal recognition setting



- agents are optimal
- actions are deterministic
- full observability (for agents and observer)
- no diagonal moves

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Goal Recognition

A goal recognition setting



	A	B	C	D	E
5	G₀				G₁
4					
3					
2					
1			Start		

- agents are optimal
- actions are deterministic
- full observability (for agents and observer)
- no diagonal moves

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Goal Recognition Design (GRD)

Offline design as a way to facilitate online goal recognition



Minimizing Worst Case Distinctiveness (WCD)

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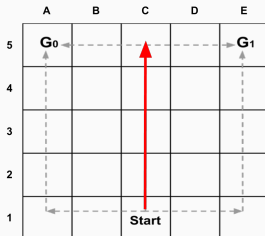
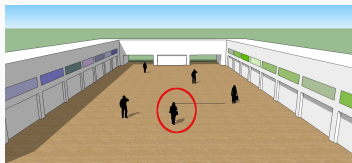
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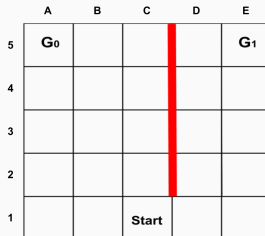
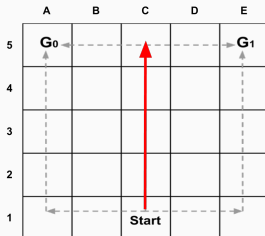
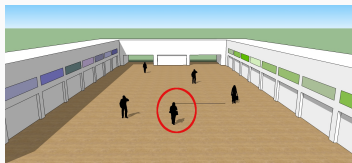
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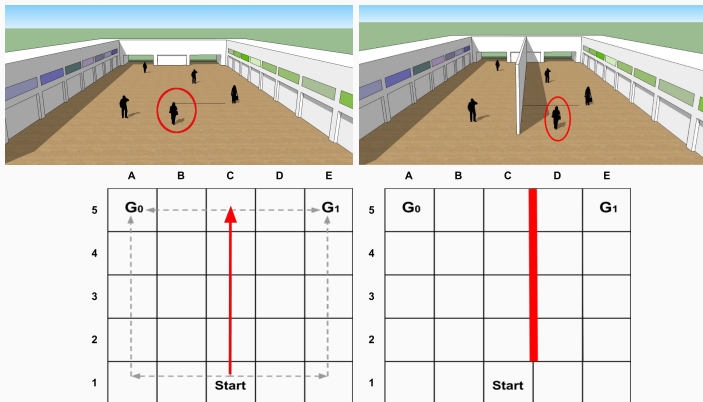
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Goal Recognition Design (GRD)

Offline design as a way to facilitate **online** goal recognition



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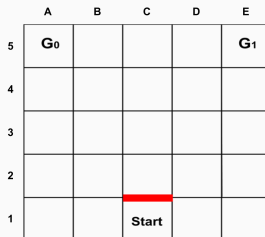
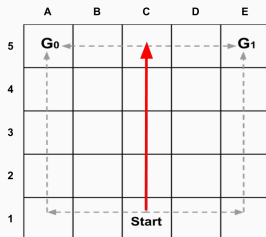
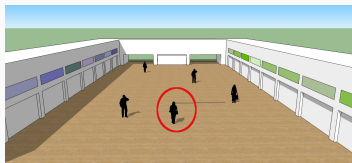
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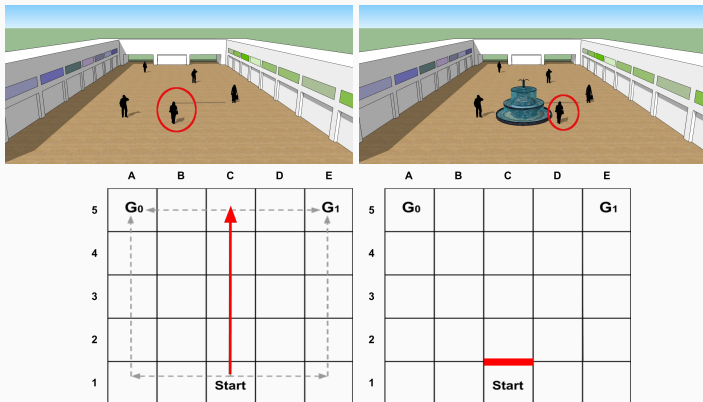
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Minimizing Worst Case Distinctiveness (WCD)

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Compilations to planning

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Compilations to planning

- Agents are assumed to be planning agents.
- We assess a model (e.g., compute WCD) by compiling the observer and actor planning problems into a single problem.

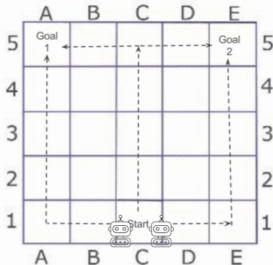
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Definition 3 For a goal recognition design problem $D = \langle P, G \rangle$ and cost function C , where $P = \langle F, I, A \rangle$ and $G = \{G_0, G_1\}$ we create a planning problem $P' = \langle F', I', A', G' \rangle$ and cost function C' that are defined as follows:

- $F' = \{f_0, f_1 | f_i \in F\} \cup \{split\} \cup \{done_0\}$
- $I' = \{f_0, f_1 | f_i \in I\}$
- $A' = \{A_{0,1}, A_0, A_1\} \cup \{DoSplit\} \cup \{Done_0\}$ where
 - $A_{0,1} = \{\{f_0, f_1 | f \in pre(a)\} \cup \{-split\}, \{f_0, f_1 | f \in add(a)\}, \{f_0, f_1 | f \in del(a)\}\} | a \in A$
 - $A_0 = \{\{f_0 | f \in pre(a)\} \cup \{split\} \cup \{\neg done_0\}, \{f_0 | f \in add(a)\}, \{f_0 | f \in del(a)\}\} | a \in A$
 - $A_1 = \{\{f_1 | f \in pre(a)\} \cup \{split\} \cup \{done_0\}, \{f_1 | f \in add(a)\}, \{f_1 | f \in del(a)\}\} | a \in A$
 - $\{DoSplit\} = \langle \emptyset, split, \emptyset \rangle$
 - $\{Done_0\} = \langle split, done_0, \emptyset \rangle$
- $G' = \langle f_0 | f \in G_0 \rangle \cup \langle f_1 | f \in G_1 \rangle$
- C' s.t. $C'(a_0) = C'(a_1) = C(a)$, $C'(a_{0,1}) = 2C(a) - \epsilon$

Inverse RL

Motivation



https://youtu.be/EcxpbhDeuZw?si=yYsM0D-hZc5_KrvG6t=329

<https://youtu.be/Z-eU5xZW7cU?si=ArJos3w273Px-d05>

<https://youtu.be/h7uGyBcIeII?si=bvAXsU2TtDTuPTkn>

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Motivation

In standard reinforcement learning, the reward function is given.

$$\text{MDP} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$$

The agent learns a policy:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right].$$

Inverse RL

In inverse reinforcement learning, we observe behavior and try to infer the reward function that explains it.

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Forward RL vs. Inverse RL

Forward RL

$$R \rightarrow \pi^*$$

- Reward is known.
- Policy is learned.
- Goal: maximize return.

Inverse RL

$$\pi_E \rightarrow R$$

- Expert behavior is observed.
- Reward is unknown.
- Goal: explain behavior.

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Problem Setup

We are given expert demonstrations:

$$\mathcal{D} = \{\tau_1, \dots, \tau_N\},$$

where each trajectory is

$$\tau = (s_0, a_0, s_1, a_1, \dots, s_T).$$

The goal is to infer a reward function:

$$R(s, a)$$

such that the expert policy appears approximately optimal.

Core question

What reward function would make the observed behavior rational?

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Why Not Just Imitate?

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Why Not Just Imitate?

Behavior cloning directly learns:

$$\pi(a \mid s)$$

from expert demonstrations.

Limitation

Behavior cloning copies actions, but does not explain why those actions were chosen.

Inverse RL tries to recover the underlying objective.

- More robust to new states.
- More transferable across environments.
- Useful when the expert objective is hard to specify manually.

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Inverse RL is ill-posed.

Many reward functions can explain the same behavior.

$$R_1 \Rightarrow \pi_E, \quad R_2 \Rightarrow \pi_E, \quad R_3 \Rightarrow \pi_E.$$

Example

A robot may avoid a corridor because it is unsafe, too long, blocked, socially inappropriate, or simply irrelevant to the task.

The demonstrations alone may not reveal which explanation is correct.

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Maximum Entropy Inverse Reinforcement Learning

Classical IRL asks:

Find a reward function R that explains the demonstrations.

However, many reward functions produce exactly the same optimal policy.

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Maximum Entropy Inverse Reinforcement Learning

Classical IRL asks:

Find a reward function R that explains the demonstrations.

However, many reward functions produce exactly the same optimal policy.

Key Idea

Rather than selecting one arbitrary reward, assume the expert behaves according to the **maximum entropy distribution** that is consistent with the demonstrations.

Maximum Entropy IRL models expert behavior as *approximately* optimal while remaining maximally uncertain about unexplained choices.

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The Maximum Entropy Principle

Among all trajectory distributions that explain the demonstrations, choose the one with the largest entropy.

Entropy of a trajectory distribution:

$$H(P) = - \sum_{\tau} P(\tau) \log P(\tau).$$

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The Maximum Entropy Principle

Among all trajectory distributions that explain the demonstrations, choose the one with the largest entropy.

Entropy of a trajectory distribution:

$$H(P) = - \sum_{\tau} P(\tau) \log P(\tau).$$

The resulting trajectory probability is

$$P(\tau) = \frac{\exp(R(\tau))}{Z},$$

where

$$R(\tau) = \sum_t R(s_t, a_t), \quad Z = \sum_{\tau} \exp(R(\tau)).$$

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The Maximum Entropy Principle

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Higher-reward trajectories become exponentially more likely, but lower-reward trajectories remain possible.

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Learning the Reward Function

Assume the reward is linear in a feature vector:

$$R(s, a) = w^\top \phi(s, a).$$

Therefore,

$$R(\tau) = w^\top \sum_t \phi(s_t, a_t).$$

Given demonstrations

$$\mathcal{D} = \{\tau_1, \dots, \tau_N\},$$

the measure is

$$\mathcal{L}(w) = \underbrace{\sum_i R_w(\tau_i)}_{\text{Reward assigned to the demonstrations}} - \underbrace{N \log Z(w)}_{\text{Normalization term}}$$

Learning IRL becomes $\max_w \mathcal{L}(w)$

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Feature Expectation Matching

The gradient of the log-likelihood is

$$\nabla_w \mathcal{L} = \mathbb{E}_{\text{expert}}[\phi] - \mathbb{E}_{P_w}[\phi].$$

Interpretation

Update the reward until the expected feature counts generated by the learned policy match those observed in the demonstrations.

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Feature Expectation Matching

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Interpretation

Update the reward until the expected feature counts generated by the learned policy match those observed in the demonstrations.

Algorithm:

- 1 Initialize reward weights.
- 2 Compute the induced policy.
- 3 Estimate expected feature counts.
- 4 Compare with demonstrations.
- 5 Update the weights.
- 6 Repeat until convergence.

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Advantages and Limitations

Maximum Entropy IRL explains demonstrations by finding the least-committed probabilistic model that still prefers high-reward behavior.

Advantages

- Probabilistic model of expert behavior.
- Handles noisy or suboptimal demonstrations.
- Avoids arbitrary reward selection.
- Convex optimization for linear rewards.
- Foundation of many modern IRL methods.

Limitations

- Computing the partition function is expensive.
- Requires a known dynamics model.
- Linear rewards may be restrictive.
- Large state spaces require approximation.

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Inverse RL is useful when the reward is difficult to write manually.

- Autonomous driving
- Robotics from demonstration
- Assistive agents
- Human-AI collaboration
- Recommender systems
- Game AI
- Modeling human decision making

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Main Challenges

- Reward ambiguity
- Limited demonstrations
- Suboptimal experts
- Distribution shift
- Computational cost of repeatedly solving RL problems
- Choosing meaningful features
- Evaluating learned rewards

Key difficulty

Good imitation does not necessarily mean that the correct reward was learned.

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IRL methods are commonly evaluated by asking:

- Does the learned policy imitate expert behavior?
- Does the learned reward generalize to new environments?
- Does the reward assign higher value to expert behavior?
- Can humans interpret the learned reward?
- Does the learned policy outperform behavior cloning under distribution shift?

- IRL learns rewards from demonstrations.
- It reverses the standard RL problem.
- The problem is fundamentally ambiguous.
- Linear IRL explains behavior through feature weights.
- MaxEnt IRL handles stochastic and noisy behavior.
- Deep IRL extends the idea to complex inputs.
- IRL is useful when specifying rewards manually is hard.

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Multi-Agent AI and MARL

We have only seen a small sample of the many challenges studied in **Multi-Agent AI** and **Multi-Agent Reinforcement Learning**.

- Cooperative, competitive, and mixed-motive environments.
- Communication, coordination, and negotiation.
- Planning, learning, and game-theoretic reasoning.
- Partial observability and uncertainty.
- Human-AI collaboration and multi-human interaction.
- Swarms, robot teams, autonomous vehicles, energy systems, and markets.
- Safety, robustness, interpretability, and scalability.



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The central challenge is no longer learning *how a single agent should act*, but how multiple intelligent agents can reason, learn, coordinate, and adapt while influencing one another.

